

FIITJEE – JEE (Main)

PHYSICS, CHEMISTRY & MATHEMATICS

QP Code: _____

Time Allotted: 3 Hours

Maximum Marks: 300

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

Important Instructions

Caution: Question Paper CODE as given above **MUST** be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & B** in the OMR.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices ext. except the Admit Card inside the examination hall / room.

B. Filling of OMR Sheet:

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.
4. **Do not fold or make any stray marks on the Answer Sheet.**

C. Marking Scheme for All Two Parts:

- (i) **Part-A (01-20)** – Contains Twenty (20) multiple choice objective questions which have four (4) options each and only one correct option. Each question carries **+4 marks** which will be awarded for every correct answer and **–1 mark** will be deducted for every incorrect answer.
- (ii) **Part-B (01-05)** contains five (05) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

BATCH – ABC LOT NAO PAPER-2123 (PHASE–5)

SECTION – I : PHYSICS

PART A: (Single Correct Answer Type)

*This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.*

1. The half life of radio active Radon is 3.8 days. The time at the end of which $(1/20)$ th of the Radon sample will remain undecayed is, [given $\log_{10} e = 0.4343$]
 (A) 13.8 days (B) 16.5 days
 (C) 53 days (D) 76 days

2. The ratio of intensities of two waves is 1:16. The ratio of maximum to minimum intensities when these waves interfere is
 (A) $\frac{25}{3}$ (B) $\frac{25}{5}$
 (C) $\frac{25}{9}$ (D) $\frac{25}{11}$

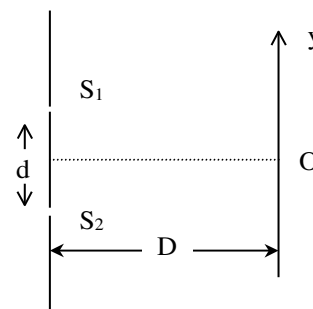
3. The cut-off wavelength when a potential difference of 25 kV is applied to an X-ray tube, is
 (A) $0.248 \text{ \AA}$ (B) $0.496 \text{ \AA}$
 (C) $0.124 \text{ \AA}$ (D) $0.620 \text{ \AA}$

4. The electron in a hydrogen atom makes a transition from an excited state to the ground state. Which of the following statements is true ?
 (A) Its kinetic energy increases and its potential and total energies decrease.
 (B) Its kinetic energy decreases, potential energy increases and its total energy remains same.
 (C) Its kinetic energy and total energy decreases and its potential energy increases.
 (D) its kinetic, potential and total energies decrease.

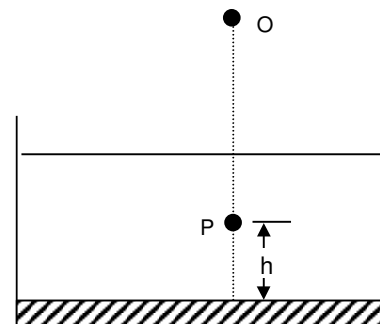
5. Electrons with energy 80 keV are incident on the tungsten target of an X-ray tube. K shell electrons of tungsten have -72.5 keV energy . X-rays emitted by the tube contain only
 (A) a continuous x-ray spectrum with a minimum wavelength of $0.155 \text{ \AA}$.
 (B) a continuous X-ray spectrum with all wavelengths.
 (C) the characteristic X-ray spectrum of tungsten.
 (D) a continuous X-ray spectrum with a minimum wavelength of $0.155 \text{ \AA}$ and the characteristic X-ray spectrum of tungsten superimposed over it.

Space for rough work

6. Consider a usual set up of Young's double slit experiment with slits illuminated with equal intensities as shown in the figure. Take 'O' as origin and the Y axis as indicated. If average intensity between $y_1 = \frac{\lambda D}{4d}$ and $y_2 = +\frac{\lambda D}{4d}$ equals 'n' times the intensity of maxima, then 'n' equals

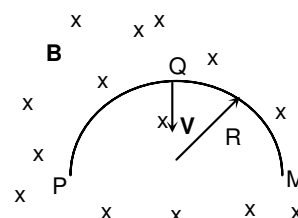
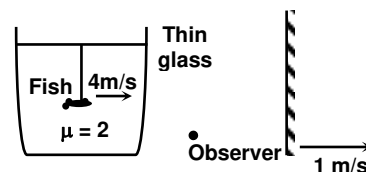
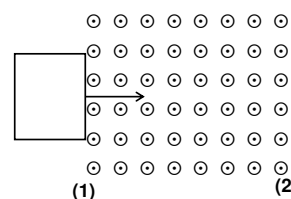


- (A) $\frac{1}{2} \left(1 + \frac{2}{\pi} \right)$ (B) $2 \left(1 + \frac{2}{\pi} \right)$
 (C) $\left(1 + \frac{2}{\pi} \right)$ (D) $\frac{1}{2} \left(1 - \frac{2}{\pi} \right)$
7. Light of wavelength λ in air enters a medium of refractive index μ . Two points in this medium, lying along the path of this light, are at distance x apart. The phase difference between these points is
- (A) $\mu \frac{2\pi}{\lambda} x$ (B) $\frac{1}{\mu} \cdot \frac{2\pi}{\lambda} x$
 (C) $(\mu - 1) \frac{2\pi}{\lambda} x$ (D) $\frac{1}{(\mu - 1)} \frac{2\pi}{\lambda} x$
8. To an eye placed at a depth h below the calm surface of refractive index μ in a lake, the surface appears like a reflecting plane mirror with a circular hole of radius R through which objects situated outside water can be seen. Then R is equal to
- (A) h (B) $\frac{h}{\mu}$
 (C) μh (D) $\frac{h}{\sqrt{\mu^2 - 1}}$
9. A plane mirror is placed at the bottom of a tank containing a liquid of refractive index μ . P is a small object at a height h above the mirror. An observer O – vertically above P, outside the liquid – sees P and its image in the mirror. The apparent distance between these two will be
- (A) $2\mu h$ (B) $\frac{2h}{\mu}$
 (C) $\frac{2h}{\mu - 1}$ (D) $h \left(1 + \frac{1}{\mu} \right)$



Space for rough work

10. The focal length of a planoconvex lens is 20 cm. The plane face of it is silvered. An object is placed at a distance of 60 cm from the lens on its axis on the convex side. Then the distance (in cm) of the image is
 (A) 12 (B) 30
 (C) 40 (D) 60
11. In a Young's double slit experiment, the fringes are displaced by a distance 'x' when a glass plate of refractive index 1.5 is introduced in the path of one of the beams. When this plate is replaced by another plate of same thickness, the shift of fringes is $(3/2)x$. The refractive index of second plate is
 (A) 1.75 (B) 1.50
 (C) 1.25 (D) 1.00
12. According to Bohr's Model of hydrogen atom
 (A) Linear velocity of electron is quantized
 (B) Linear momentum of electron is quantized
 (C) Angular velocity of electron is quantized
 (D) Angular momentum of electron is quantized
13. A loop of area A enters a region of magnetic field at position -1 and it is leaving the field at position -2. The direction of induced current in the two cases will be
 (A) anticlockwise, clockwise.
 (B) clockwise, anticlockwise
 (C) clockwise, clockwise.
 (D) anticlockwise, anticlockwise
14. What is the velocity of image of fish in the mirror with respect to ground is
 (A) 2 m/s (B) 4 m/s
 (C) 5 m/s (D) zero
15. A conducting thin semicircular ring of radius R falls in a horizontal magnetic induction B such that $\vec{v} \cdot \vec{B} = 0$. At the position PQM, the speed of ring is v. The potential difference developed across the ring is
 (A) πRBv , M at higher potential
 (B) $2RBv$, M at higher potential
 (C) zero
 (D) $2RBv$, P at higher potential

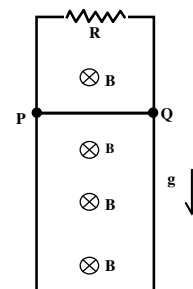


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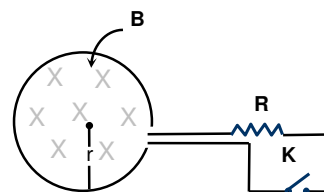
16. An ideal coil of 10H is connected in series with a resistance of $5\ \Omega$ and a battery of 5V . 2 second after the connection is made the current flowing in amperes in the circuit is
 (A) $(1 - e)$ (B) e
 (C) e^{-1} (D) $(1 - e^{-1})$

17. A coil of inductance 300 mH and resistance $2\ \Omega$ is connected to a source of voltage 2V . The current reaches half of its steady state value in
 (A) 0.05 s (B) 0.1 s
 (C) 0.15 s (D) 0.3 s

18. A rod PQ mass ' m ' and length ' ℓ ' can slide without friction on two vertical conducting semi infinite rails. It is given a velocity V_0 downwards, so that it continues to move downward with the same speed V_0 on its own at any later instant of time. Assuming g to be constant every where, the value of $\frac{V_0 B^2 L^2}{mgR}$ is
 (A) 1 (B) 2
 (C) 3 (D) 4



19. Shown in the figure is a circular loop of radius r and resistance R . A variable magnetic field of induction $B = e^{-t}$ is established inside the coil. If the key (K) is closed at $t = 0$, the electrical power developed at the instant is equal to
 (A) $\frac{\pi r^2}{R}$ (B) $\frac{10r^3}{R}$
 (C) $\frac{\pi^2 r^4 R}{5}$ (D) $\frac{\pi^2 r^4}{4R}$



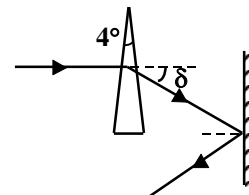
20. A thin circular ring of area A is held perpendicular to a uniform magnetic field of induction B . A small cut is made in the ring and a galvanometer is connected across the ends such that the total resistance of the circuit is R . When the ring is suddenly squeezed to zero area, the charge flowing through the galvanometer is
 (A) $\frac{BR}{A}$ (B) $\frac{AB}{R}$
 (C) ABR (D) $\frac{B^2 A}{R^2}$

Space for rough work

PART – B: (Integer Answer Type)

*This section contains **05 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.*

1. A horizontal ray of light passes through a prism of $\mu = 1.5$ whose apex angle is 4° and then strikes a vertical mirror M as shown. For the ray after reflection to become horizontal, the mirror must be rotated through an angle of x° . Find x .



2. For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Find the refractive index of the prism.
3. The index of refraction of diamond is 2.0. Velocity of light in diamond is $\frac{k}{2} \times 10^{10}$ cm/sec (approximately). Find the value of k .
4. Two radioactive materials X_1 and X_2 have decay constants 10λ and λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of X_1 to that of X_2 will be $1/e$ after a time $\frac{1}{n\lambda}$. Find n .
5. The flux linked with a coil at any instant 't' is given by $\phi = 10t^2 - 50t + 250$. Find the magnitude of induced emf at $t = 3$ s is 2k volts. Find the value of k .

Space for rough work

SECTION - II : CHEMISTRY**PART A: (Single Correct Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

- One coulomb of charge passes through solution of AgNO_3 and CuSO_4 connected in series and the conc. of two solutions being in the ratio 1:2. The ratio of weight of Ag and Cu deposited on Pt electrode is
(A) 107.9 : 63.54 (B) 54 : 31.77
(C) 107.9 : 31.77 (D) 54 : 63.54
- The amount of ice that will separate on cooling a solution containing 50g of ethylene glycol in 200g water to -9.3°C is : [$K_f = 1.86 \text{ K molality}^{-1}$]
(A) 38.71 g (B) 38.71 mg
(C) 42 g (D) 42 mg
- To remove the acidic impurity, the flux that can be used is a
(A) SiO_2 (B) CaO
(C) NaCl (D) CuO
- Dettol consist of
(A) cresol + ethanol (B) xlenol + terpinol
(C) chloroxlenol + terpeneol (D) none of these
- The least stable hydride of 15th group elements is
(A) NH_3 (B) PH_3
(C) AsH_3 (D) SbH_3
- For the following cell,
 $\text{Zn}(s) | \text{ZnSO}_4(aq) || \text{CuSO}_4(aq) | \text{Cu}(s)$
When the concentration of Zn^{2+} is 10 times the concentration of Cu^{2+} , the expression for ΔG (in J mol^{-1}) is
[F is Faraday constant; R is gas constant; T is temperature; $E^0(\text{cell}) = 1.1 \text{ V}$]
(A) 1.1 F (B) $2.303 RT - 2.2 F$
(C) $2.303 RT + 1.1 F$ (D) $-2.2 F$

Space for rough work

7. Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 sol is
 (A) Na_2SO_4 (B) CaCl_2
 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) NH_4Cl
8. The freezing point (in $^\circ\text{C}$) of a solution containing 0.1 g of $\text{K}_3[\text{Fe}(\text{CN})_6]$ (Mol. Wt. 329) in 100 g of water ($K_f = 1.86 \text{ K kg mol}^{-1}$) is
 (A) -2.3×10^{-2} (B) -5.7×10^{-2}
 (C) -5.7×10^{-3} (D) -1.2×10^{-2}
9. Given the following half reactions,
 $\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s}) \quad E^\circ = +0.799 \text{ V}$
 $\text{AgCl}(\text{s}) + \text{e}^- \longrightarrow \text{Ag}(\text{s}) + \text{Cl}^-(\text{aq}) \quad E^\circ = +0.222 \text{ V}$
 What is the molar solubility of AgCl ? [Given antilog $(-9.758) = 1.743 \times 10^{-10}$]
 (A) $4.46 \times 10^{-9} \text{ M}$ (B) $1.32 \times 10^{-5} \text{ M}$
 (C) $1.79 \times 10^{-10} \text{ M}$ (D) 0.577 M
10. The boiling point of an aqueous solution of a non-volatile solute is 100.15°C . What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water? The values of K_b and K_f for water are $0.512 \text{ K kg mol}^{-1}$ and $1.86 \text{ K kg mol}^{-1}$ respectively.
 (A) -0.544°C (B) -0.512°C
 (C) -0.272°C (D) -1.86°C
11. On passing H_2S gas into the aqueous solution of an inorganic salt gives a black precipitate which dissolves in aqua-regia. The solution is evaporated to dryness and residue is extracted with dilute HCl . When hot nitrite solution is added along with the excess of acetic acid into this solution, a yellow coloured precipitate is obtained. Hence the inorganic salt may be:
 (A) NiCl_2 (B) $\text{Hg}(\text{NO}_3)_2$
 (C) CoCl_2 (D) MnCl_2
12. A face centred cubic lattice is made up of hollow spheres of X. Spheres of solid Y are present in hollow sphere X. Radius of Y is half of radius of X. What is the ratio of total volume of spheres of X unoccupied by Y in a Unit-cell and volume of unit cell?
 (A) $\frac{7\pi}{24\sqrt{2}}$ (B) $\frac{7\pi}{24}$
 (C) $\frac{7\sqrt{2}\pi}{24}$ (D) None of these

Space for rough work

13. During extraction of Fe
 - (A) Carbon is ultimately converted to CO_2 in combustion zone in blast furnace.
 - (B) CO_2 is one of the by-products of reduction zone of blast furnace.
 - (C) Cast iron is obtained from blast furnace.
 - (D) Uppermost zone of blast furnace is slag formation zone.

14. Which among the following statement is incorrect?
 - (A) With excess ammonia, chlorine gives N_2 and NH_4Cl .
 - (B) With hot and conc. NaOH , Cl_2 gives a mixture of NaClO_3 , NaCl and H_2O .
 - (C) HCl reacts with finely powdered iron to form ferric chloride.
 - (D) Bond dissociation energy of $\text{F}-\text{F}$ is less than that of $\text{Cl}-\text{Cl}$.

15. Which of the following are diamagnetic?
 1. $\text{K}_4[\text{Fe}(\text{CN})_6]$
 2. $\text{K}_3[\text{Cr}(\text{CN})_6]$
 3. $\text{K}_3[\text{Co}(\text{CN})_6]$
 4. $\text{K}_2[\text{Ni}(\text{CN})_4]$
 Select the correct choice using the codes given below

(A) 1, 2 and 4	(B) 1, 3 and 4
(C) 2 and 3	(D) 1 and 4

16. A compound (A) forms an unstable pale blue colour solution in water. The solution decolourizes Br_2 water and an acidified solution of KMnO_4 . The compound (A) is

(A) HNO_2	(B) HNO_3
(C) N_2O_5	(D) None of the above

17. The vapour pressure lowering of a 0.1 M aqueous solution of non-electrolyte at 75°C is (ΔH_v° of water = 9720 cal/mol and antilog 0.407 = 2.55)

(A) 0.53 torr	(B) 0.78 torr
(C) 0.02 torr	(D) 0.22 torr

18. Which of the following shall form an octahedral complex?

(A) d^4 (low spin)	(B) d^8 (high spin)
(C) d^6 (high spin)	(D) none of these

19. Which type of 'defect' has the presence of cations in the interstitial sites?

(A) Metal deficiency defect	(B) Schottky defect
(C) Vacancy defect	(D) Frenkel defect

20. Which of the vitamins given below is water soluble?

(A) Vitamin D	(B) Vitamin E
(C) Vitamin K	(D) Vitamin C

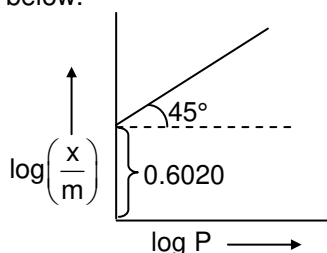
Space for rough work

PART – B: (Integer Answer Type)

*This section contains **05 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.*

1. How many of following compounds will react with excess of KCN to form complex ions?
 AgNO_3 , PbNO_3 , CuS , CuCN , $\text{Cd}(\text{CN})_2$, $\text{Fe}(\text{CN})_2$, $\text{Fe}(\text{CN})_3$, CoCl_2 , NiCl_2 , $\text{Ni}(\text{CN})_2$
2. 1.2 kg of ethylene glycol was added to 9 litre water contained in a vessel. The solution is just prevented from freezing at -4°C . Due to a sudden drop in temperature to -6°C , what will be the mass of ice separated out in kilograms. $K_f \text{ H}_2\text{O} = 1.86 \text{ K Kg mol}^{-1}$. (Assume the density of water does not change)
3. A weak field complex of Ni^{2+} has magnetic moment value of 2.82 BM. The number of electrons in the t_{2g} level of Ni^{2+} will be.....
4. A complex of potassium, iron and cyanide ions $\text{K}_x[\text{Fe}(\text{CN})_6]$ is 100% ionized at 1 molal. If its elevation in boiling points is 2.08°C , find out the value of 'x' in the complex.
 (Given $K_b = 0.52^\circ\text{C mol}^{-1} \text{ kg}$)

5. Graph between $\log\left(\frac{x}{m}\right)$ and $\log p$ is straight line at angle of 45° with the intercept of 0.6020 as shown below:



Calculate the extent of adsorption $\left(\frac{x}{m}\right)$ at a pressure of 1 atm.

Space for rough work

SECTION - III : MATHEMATICS

PART - A: (Single Correct Answer Type)

This section contains 20 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. If A is a matrix such that $A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $A^2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, then sum of square of elements in A is
 (A) 21 (B) 3
 (C) 6 (D) none of these
2. If $\hat{a}, \hat{b}, \hat{c}$ are three non-coplanar unit vectors, then $[\hat{a} \ \bar{p} \ \bar{q}]\hat{a} + [\hat{b} \ \bar{p} \ \bar{q}]\hat{b} + [\hat{c} \ \bar{p} \ \bar{q}]\hat{c}$ is equal to
 (A) $(\hat{a} + \hat{b} + \hat{c}) \times (\bar{p} \times \bar{q})$ (B) $\hat{a} + \hat{b} + \hat{c} + \bar{p} + \bar{q}$
 (C) $\bar{p} + \bar{q}$ (D) $\bar{p} \times \bar{q}$
3. If L_1 is the line of intersection of the planes $2x - 2y + 3z - 2 = 0$, $x - y + z + 1 = 0$ and L_2 is the line of intersection of the planes $x + 2y - z - 3 = 0$, $3x - y + 2z - 1 = 0$, then the distance of the origin from the plane, containing the lines L_1 and L_2 , is :
 (A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{4\sqrt{2}}$
 (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$
4. The area (in sq. units) of the region $\{(x, y) : y^2 \geq 2x \text{ and } x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$ is:
 (A) $\pi - \frac{8}{3}$ (B) $\pi - \frac{4\sqrt{2}}{3}$
 (C) $\frac{\pi}{2} - \frac{2\sqrt{2}}{3}$ (D) $\pi - \frac{4}{3}$

Space for rough work

5. If the system of linear equations
 $x + ky + 3z = 0$
 $3x + ky - 2z = 0$
 $2x + 4y - 3z = 0$
 has a non-zero solution (x, y, z) , then $\frac{xz}{y^2}$ is equal to :
- (A) 30 (B) -10
 (C) 10 (D) -30
6. At present, a firm is manufacturing 2000 items. It is estimated that the rate of change of production P w.r.t. additional number of workers x is given by $\frac{dP}{dx} = 100 - 12\sqrt{x}$. If the firm employs 25 more workers, then the new level of production of items is
- (A) 3000 (B) 3500
 (C) 4500 (D) 2500
7. Let A be a square matrix all of whose entries are integers. Then which one of the following is true?
- (A) If $\det A = \pm 1$, then A^{-1} exists but all its entries are not necessarily integers
 (B) If $\det A \neq \pm 1$, then A^{-1} exists and all its entries are non-integers
 (C) If $\det A = \pm 1$, then A^{-1} exists and all its entries are integers
 (D) If $\det A = \pm 1$, then A^{-1} need not exist
8. A and B play a game of tennis. The situation of the game is as follows; if one scores two consecutive points after a deuce he wins; if loss of a point is followed by win of a point, it is deuce. The chance of a server to win a point is $\frac{2}{3}$. The game is at deuce and A is serving. Probability that A will lose the game is, (serves are changed after each point)
- (A) $\frac{3}{5}$ (B) $\frac{2}{5}$
 (C) $\frac{1}{2}$ (D) $\frac{4}{5}$
9. The equation of the plane bisecting the acute angle between the planes $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is
- (A) $23x - 13y + 32z + 45 = 0$ (B) $5x - y - 4z = 3$
 (C) $5x - y - 4z + 45 = 0$ (D) $23x - 13y + 32z + 3 = 0$
10. The equation of the sphere concentric with the sphere $x^2 + y^2 + z^2 - 4x - 6y - 8z - 5 = 0$ and which passes through the point $(0, 1, 0)$ is
- (A) $x^2 + y^2 + z^2 + 4x + 6y - 8z - 7 = 0$ (B) $x^2 + y^2 + z^2 - 4x - 6y - 8z = 0$
 (C) $x^2 + y^2 + z^2 - 4x - 6y - 8z + 5 = 0$ (D) None of these

Space for rough work

11. The area bounded by $y = \cos^{-1}(\cos x)$, x -axis, $0 \leq x \leq 2\pi$, is
 (A) π (B) 2π
 (C) π^2 (D) $2\pi^2$
12. If $a, b, c > 0$ and $x, y, z \in \mathbb{R}$, then the determinant $\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$ is equal to
 (A) $a^x + b^y + c^z$ (B) $a^{-x} b^{-y} c^{-z}$
 (C) $a^{2x} b^{2y} c^{2z}$ (D) 0
13. A die is thrown n times (n being odd). The probability that even face turns odd number of times, is
 (A) $\frac{n}{2n+1}$ (B) less than $\frac{1}{2}$
 (C) $\frac{1}{2}$ (D) greater than $\frac{1}{2}$
14. The degree and order of the differential equation of all the parabolas whose axis is x -axis are
 (A) 2, 1 (B) 1, 2
 (C) 3, 2 (D) none of these
15. If the matrix $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal, then
 (A) $\alpha = \pm \frac{1}{\sqrt{2}}$ (B) $\beta = \pm \frac{1}{\sqrt{6}}$
 (C) $\gamma = \pm \frac{1}{\sqrt{3}}$ (D) all of these

Space for rough work

16. Consider $\triangle ABC$ and $\triangle A_1B_1C_1$ in such a way that $\overrightarrow{AB} = \overrightarrow{A_1B_1}$ and M, N, M₁, N₁ be the mid points of AB, BC, A₁B₁ and B₁C₁ respectively, then
- (A) $\overrightarrow{MM_1} = \overrightarrow{NN_1}$ (B) $\overrightarrow{CC_1} = \overrightarrow{MM_1}$
 (C) $\overrightarrow{CC_1} = \overrightarrow{NN_1}$ (D) $\overrightarrow{MM_1} = \overrightarrow{BB_1}$
17. The length of perpendicular from the point $(-1, 2, -2)$ on the line $\frac{x+1}{2} = \frac{y-2}{-3} = \frac{z+2}{4}$ is
- (A) $\sqrt{29}$ (B) $\sqrt{6}$
 (C) $\sqrt{21}$ (D) none of these
18. Let m be a positive integer and
- $$\Delta_r = \begin{vmatrix} 2r-1 & {}^m C_r & 1 \\ m^2-1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix}$$
- Then the value of $\sum_{r=0}^m \Delta_r$ is given by
- (A) 0 (B) m^2-1
 (C) 2^m (D) $2^m \sin^2(2^m)$
19. Let A, B, C be three events such that A and B are independent and $P(C) = 0$, then events A, B, C are
- (A) independent
 (B) pairwise independent but not totally independent
 (C) $P(A) = P(B) = P(C)$
 (D) none of these
20. The curve, which satisfies the differential equation $\frac{xdy - ydx}{xdy + ydx} = y^2 \sin(xy)$ and passes through $(0, 1)$, is given by
- (A) $y(1 - \cos xy) + x = 0$ (B) $\sin xy - x = 0$
 (C) $\sin y + y = 0$ (D) $\cos xy - 2y = 0$

Space for rough work

PART – B: (Integer Answer Type)

This section contains **05 questions**. Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

1. If 'a' is a real constant ($|a| \geq 2$) and A, B, C are variable angles and $\sqrt{a^2 - 4} \tan A + a \tan B + \sqrt{a^2 + 4} \tan C = 6a$, then the least value of $\frac{1}{3} (\tan^2 A + \tan^2 B + \tan^2 C)$ is _____
 2. The degree of the differential equation of all tangent lines to the parabola $y = x^2$ is _____
 3. The maximum area of a rectangle whose two vertices lie on the x-axis and two on the curve $y = 3 - |x|$, $-3 \leq x \leq 3$ is k, then 2k is equal to _____
 4. The area enclosed between the curves $y = \sin^2 x$ and $y = \cos^2 x$ in the interval $0 \leq x \leq \pi$ is p, then 4p is equal to _____
 5. Entries of a 2×2 determinant are chosen from the set $\{-1, 1\}$. The probability that determinant has zero value is p, then 6p is equal to _____
-

Space for rough work

FIITJEE INTERNAL TEST

PHYSICS, CHEMISTRY & MATHEMATICS

CPT-2

QP CODE:

ANSWERS

PHYSICS (SECTION -I)

PART-A

- | | | | |
|-------|-------|-------|-------|
| 1. B | 2. C | 3. B | 4. A |
| 5. D | 6. A | 7. A | 8. D |
| 9. B | 10. A | 11. A | 12. D |
| 13. B | 14. D | 15. B | 16. D |
| 17. B | 18. A | 19. D | 20. B |

PART-B

- | | | | |
|------|------|------|------|
| 1. 1 | 2. 2 | 3. 3 | 4. 9 |
| 5. 5 | | | |

CHEMISTRY (SECTION -II)

PART-A

- | | | | |
|-------|-------|-------|-------|
| 1. C | 2. A | 3. B | 4. C |
| 5. D | 6. B | 7. C | 8. A |
| 9. B | 10. C | 11. C | 12. A |
| 13. B | 14. C | 15. A | 16. A |
| 17. A | 18. A | 19. D | 20. D |

PART-B

- | | | | |
|------|------|------|------|
| 1. 9 | 2. 3 | 3. 6 | 4. 3 |
| 5. 4 | | | |

MATHEMATICS (SECTION-III)

PART - A

- | | | | |
|-------|-------|-------|-------|
| 1. A | 2. D | 3. C | 4. A |
| 5. C | 6. B | 7. C | 8. C |
| 9. A | 10. C | 11. C | 12. D |
| 13. C | 14. B | 15. D | 16. D |
| 17. D | 18. A | 19. A | 20. A |

PART - B

- | | | | |
|---------|---------|---------|---------|
| 1. 4.00 | 2. 2.00 | 3. 9.00 | 4. 4.00 |
| 5. 3.00 | | | |

SOLUTION

PHYSICS

PART-A

7. $\phi = \frac{2\pi}{\lambda} \times \Delta x$

$$\lambda_m = \frac{\lambda}{\mu}$$

$$\phi = \frac{2\pi}{\lambda_m} \times \Delta x = \frac{2\pi}{\lambda_0 / \mu} \times \Delta x$$

$$\phi = \mu \frac{2\pi}{\lambda_0} \times \Delta x \text{ and } \Delta x = x$$

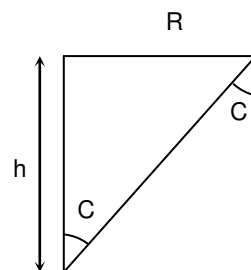
$$\phi = \mu \frac{2\pi}{\lambda} x$$

8. $\sin C = \frac{1}{\mu}$

$$\sin C = \frac{1}{\mu} = \frac{P}{H}$$

$$b = \sqrt{\mu^2 - 1}$$

$$\tan C = \frac{1}{\sqrt{\mu^2 - 1}}$$



9. Let point P is at d from surface

$$d_{app} = \frac{d}{\mu}$$

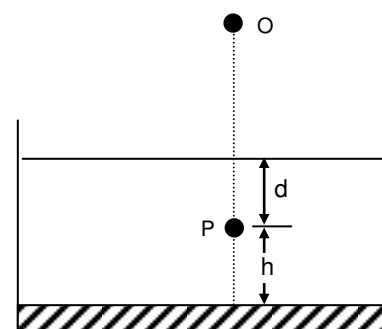
...(i)

Image by mirror will be at h beyond mirror

$$d_{App} = \frac{d + h + h}{\mu} = \frac{d + 2h}{\mu}$$

Distance between both will be

$$\frac{d + 2h}{\mu} - \frac{d}{\mu} = \frac{2h}{\mu}$$



10. $P_m = 2P_L + P_m$

$$-\frac{1}{f_m} = \frac{2}{f_2} - \frac{1}{f_m}$$

$$R \rightarrow \infty$$

$$\frac{1}{f} \rightarrow 0$$

$$f_m = -\frac{f_L}{2}$$

$$f_m = -10 \text{ cm}$$

$$v = \frac{uf}{u-f} = \frac{-60 \times -10}{-60+10}$$

$$v = \frac{600}{-50} = -12 \text{ cm}$$

11. $\Delta x = t(\mu - 1)$

$$\Delta x \propto (\mu - 1)$$

$$y = \frac{D}{d} \Delta x, y \propto \Delta x$$

$$y \propto (\mu - 1)$$

$$\frac{y_1}{y_2} = \frac{\mu_1 - 1}{\mu_2 - 1}$$

$$\frac{x}{\frac{3}{2}x} = \frac{1.5 - 1}{\mu_2 - 1}$$

$$\frac{2}{3} = \frac{1.5 - 1}{\mu_2 - 1}$$

$$2\mu_2 - 2 = 4.5 - 3$$

$$2\mu_2 - 2 = 1.5$$

$$\mu_2 = \frac{3.5}{2} = 1.75$$

16. $I_t = I_0(1 - e^{-t/t_c})$

$$t_c = \frac{L}{R} = \frac{10}{5} = 2$$

$$I_0 = \frac{V_0}{R}$$

$$I_0 = 5/5 = 1A$$

$$I_t = 1 \left(1 - e^{-\frac{2}{2}} \right) = 1 - \frac{1}{e}$$

$$\text{or } I_t = 1 - e^{-1}$$

18. $a = 0$

$$F_m - mg = 0$$

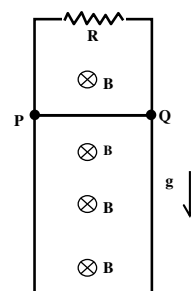
$$F_m = mg$$

$$i\ell B = mg$$

$$\Rightarrow \frac{Bv\ell(B)}{R} = mg$$

$$\frac{B^2 \ell^2 V_0}{R} = mg$$

$$\frac{V_0 B^2 L^2}{MgR} = 1$$



20. $\Delta q = \frac{\Delta \phi}{R}$

$$\Delta \phi = AB$$

$$\Delta q = \frac{AB}{R}$$

PART C :

2. $r_1 + r_2 = A$

$$r_1 = r_2 = C$$

$$2C = A \Rightarrow C = A/2$$

$$\sin C = 1/\mu \Rightarrow \mu = 2$$

4. $N \propto (e^{-\lambda t})$

$$\frac{N_1}{N_2} = \frac{e^{-\lambda_1 t}}{e^{-\lambda_2 t}}$$

$$e^{-10\lambda t} \times e = e^{-\lambda t}$$

$$e^{-10\lambda t + 1} = e^{-\lambda t}$$

$$-10\lambda t + 1 = -\lambda t$$

$$-9\lambda t = -1$$

$$\Rightarrow t = \frac{1}{9\lambda}$$

5. $\phi = 10t^2 - 50t + 250$

$$e = \left| \frac{d\phi}{dt} \right|, e = 20t - 50 \text{ at } t = 3 \text{ sec}$$

$$e = 20 \times 3 - 50 = 10 \text{ V}$$

flux is increasing at $t = 3$ sec. so emf will be produce i which is try to compensate change of ϕ .

CHEMISTRY

PART-A

1. Faraday's 2nd Law $\frac{W}{E} = \text{constant}$

$$\text{So, } \frac{W_{\text{Ag}}}{E_{\text{Ag}}} = \frac{W_{\text{Cu}}}{E_{\text{Cu}}} \quad (\text{Ag}^+ + e^- \rightarrow \text{Ag}, E_{\text{Ag}} = \frac{M}{1})$$

$$\therefore \frac{W_{\text{Ag}}}{W_{\text{Cu}}} = \frac{E_{\text{Ag}}}{E_{\text{Cu}}} = \frac{107.9}{63.54/2} \quad (\text{Cu}^{2+} + 2e^- \rightarrow \text{Cu}, E_{\text{Cu}} = \frac{M}{2})$$

$$= \frac{107.9}{31.77}$$

Hence, **(C)** is correct.

2. $T = \frac{1000 \times K_f \times w}{W \times M}$

$$\text{or } 9.3 = \frac{1000 \times 1.86 \times 50}{62 \times W}$$

$$\therefore W = 161.29$$

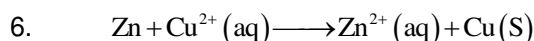
$$\therefore \text{Ice separated} = 200 - 161.29 = 38.71 \text{ g}$$

Hence, **(A)** is correct.

3. To remove acidic impurity, the flux can be used should be basic

Hence, **(B)** is correct.

5. Thermal stability of hydrides of nitrogen family decreases gradually from NH_3 to BiH_3 .
Hence, **(D)** is correct answer.



$$\Delta G = \Delta G^0 + 2.303 RT \log_{10} Q$$

$$Q = \frac{[\text{Zn}^{2+}]}{[\text{Cu}^{2+}]}$$

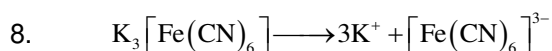
$$\Delta G^0 = -nE^0F$$

$$= -2 \times F \times 1.1$$

$$\text{Given } [\text{Zn}^{2+}] = 10[\text{Cu}^{2+}]$$

$$= 2.303RT - 2.2 F$$

7. Greater the charge on opposite charged ion greater is its coagulating power.



$$i = 4$$

Now,

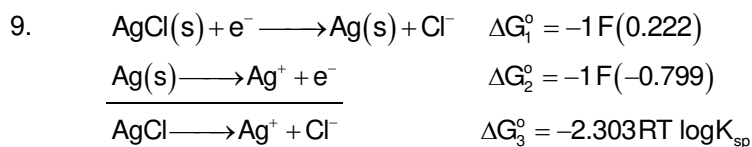
$$\Delta T_f = k_f \times i \times \text{molality}$$

$$= 1.86 \times 4 \times \frac{0.1}{329} \times \frac{1000}{100}$$

$$= 2.3 \times 10^{-2}$$

$$T_f = 0 - 2.3 \times 10^{-2}$$

$$= -2.3 \times 10^{-2} ^\circ\text{C}$$



$$\text{Also } \Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$$

We can solve for K_{sp} and solubility = $\sqrt{K_{sp}}$

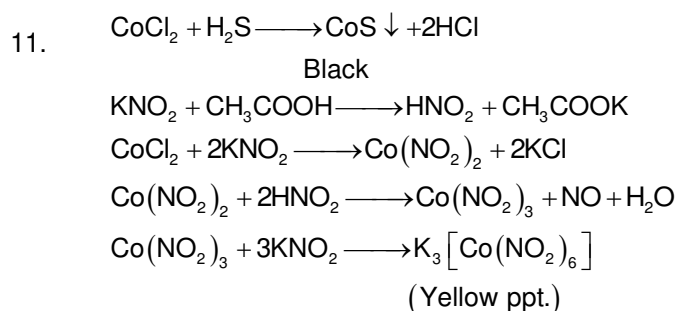
10. For same solution, $\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b}$
 or $\Delta T_f = \frac{K_f}{K_b} \times \Delta T_b = \frac{1.86}{0.512} \times 0.15 = 0.545$

Now, on diluting the solution to double

$$\Delta T_f \propto \frac{1}{\text{wt. of solvent}}$$

$$\therefore \Delta T_f = \frac{0.545}{2} = 0.272$$

$$\therefore \text{Freezing point} = -0.272 ^\circ\text{C}.$$



Hence (C) is the correct answer.

12. Effective no. of atoms of X present in a unit cell = 4

$$\text{Total volume of X unoccupied by Y in a unit cell} = 4 \times \frac{4}{3} (\text{R}^3 - r^3) \times \pi$$

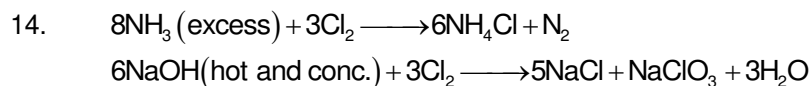
$$= \frac{14\pi\text{R}^3}{3} \left(\because r = \frac{\text{R}}{2} \right)$$

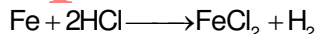
$$\text{Volume of unit cell} = a^3$$

$$\left(\frac{4\text{R}}{\sqrt{2}} \right)^3 = \frac{64}{2\sqrt{2}} \text{R}^3$$

$$\text{Derived ratio} = \frac{\frac{14\pi\text{R}^3}{3}}{\frac{64}{2\sqrt{2}} \text{R}^3} = \frac{7\pi}{24\sqrt{2}}$$

13. In blast furnace,
 In combustion zone, CO is produced ultimately which reduce Fe_2O_3 to Fe_3O_4 in reduction zone, which is uppermost zone.





Liberation of H_2 prevents the formation of FeCl_3 .
Bond dissociation of $\text{F}-\text{F}$ is less than that of $\text{Cl}-\text{Cl}$.

15. $\text{K}_4[\text{Fe}(\text{CN})_6]$, $\text{K}_3[\text{Co}(\text{CN})_6]$ and $\text{K}_2[\text{Ni}(\text{CN})_4]$
All the electrons are paired. These complexes are low spin inner orbital complex.
So they are diamagnetic complex.
16. HNO_2 is pale blue colour due to dissociation of N_2O_5 in concentrated solution. It decomposes as

$$2\text{HNO}_2 \longrightarrow \text{NO} + \text{NO}_2 + \text{H}_2\text{O}$$
 and in dilute solution it decomposes as

$$3\text{HNO}_2 \longrightarrow \text{HNO}_3 + \text{H}_2\text{O} + 2\text{NO}$$
 Reaction with (Br_2 water)
 (a) $\text{HNO}_2 + \text{H}_2\text{O} + \text{Br}_2 \longrightarrow \text{HNO}_3 + 2\text{HBr}$
 (b) $\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 + 5\text{HNO}_2 \longrightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 3\text{H}_2\text{O} + 5\text{HNO}_3$
17.
$$\log \frac{P_2}{P_1} = \frac{\Delta H^\circ}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

Vapour pressure of water at $100^\circ\text{C} = 1 \text{ atm}$
18. d^4 low spin complex will have two d-orbitals vacant required for d^2sp^3 hybridization to give an octahedral complex
19. In Frenkel defect, smaller ion displaces from its actual lattice site into the interstitial sites.
20. Vitamin C is water soluble.

PART – B

1. $\text{AgNO}_3 + \text{KCN} \longrightarrow \text{AgCN} \downarrow$
 $\text{AgCN} + \text{KCN} \longrightarrow \text{Ag}(\text{CN})_2^-$
 $2\text{CuS} + 8\text{CN}^- \longrightarrow 2[\text{Cu}(\text{CN})_4]^{3-} + \text{S}_2^{2-}$
 $\text{CuCN} + 3\text{CN}^- \longrightarrow [\text{Cu}(\text{CN})_4]^{3-}$
 $\text{CdCl}_2 + 2\text{CN}^- \longrightarrow \text{Cd}(\text{CN})_2$
 $\text{Cd}(\text{CN})_2 + 2\text{CN}^- \longrightarrow [\text{Cd}(\text{CN})_4]^{2-}$
 $\text{Fe}(\text{CN})_2 + 4\text{CN}^- \longrightarrow [\text{Fe}(\text{CN})_6]^{4-}$
 $\text{Fe}(\text{CN})_3 + 3\text{CN}^- \longrightarrow [\text{Fe}(\text{CN})_6]^{3-}$
 $\text{CoCl}_2 + 2\text{CN}^- \longrightarrow \text{Co}(\text{CN})_2$
 $\text{Co}(\text{CN})_2 + 4\text{CN}^- \longrightarrow [\text{Co}(\text{CN})_6]^{4-}$
 $\text{NiCl}_2 + 2\text{KCN} \longrightarrow \text{Ni}(\text{CN})_2$
 $\text{Ni}(\text{CN})_2 + 2\text{KCN} \longrightarrow [\text{Ni}(\text{CN})_4]^{2-}$
 $\text{Pb}^{+2} + 2\text{CN}^- \longrightarrow \text{Pb}(\text{CN})_2 \downarrow$
 $\text{Pb}(\text{CN})_2$ is not soluble in excess of KCN .
2. At -6°C , calculate the amount of water present. The difference in mass of water initially and finally is the mass of ice separated out.

$$\Delta T_f = K_f \cdot m \Rightarrow 6 = 1.86 \times \frac{1.2 \times 1000}{62 \times W_{\text{H}_2\text{O}}}$$

$$\Rightarrow W_{\text{H}_2\text{O}} = 6 \text{ kg}$$

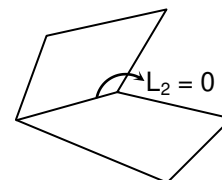
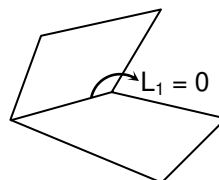
$$\therefore \text{Mass of ice separated out} = 9 - 6 = 3 \text{ kg}$$

3. $\mu = 2.82 = \sqrt{n(n+2)}$
 $\therefore n = 2$
 As 2 unpaired electrons are present, i.e. Ni^{2+} configuration is $2d^8$. The number of electrons on t_{2g} are six.
4. $\Delta T_b = K_b \times m \times i$
 $2.08 = 0.52 \times 1 \times i$
 $i = 4$
 $\text{K}_3[\text{Fe}^{III}(\text{CN})_6]$
5. $\log\left(\frac{x}{m}\right) = \log k + \frac{1}{n} \log p$
 $\log\left(\frac{x}{m}\right) = 0.6020 + 0 \quad (\because \text{Slope} = \tan 45^\circ = 1)$
 $\frac{x}{m} = \text{antilog}(0.6020)$
 $= 4$

MATHEMATICS

Part – A

1. **A**
 Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ we get four equations and four variables
 $a - b = -1$
 $c - d = 2$
 Also $A \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
 $\Rightarrow -a + 2b = 1, -c + 2d = 0$
 $a = -1, b = 0, c = 4, d = 2$
2. **D**
 $[\hat{a} \vec{p} \vec{q}] = \text{projection of } \vec{p} \times \vec{q} \text{ in the direction of } \hat{a}$. Hence the given vector is $\vec{p} \times \vec{q}$
3. **C**
 Vectors along the given lines L_1, L_2 are
 $\hat{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 3 \\ 1 & -1 & 1 \end{vmatrix} = \hat{i} + \hat{j}$
 and $\hat{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & -1 & 2 \end{vmatrix} = 3\hat{i} - 5\hat{j} - 7\hat{k}$
 Putting $y = 0$ in 1st two equation
 $2x + 3z = 2$
 $2x + 2z = -2$
 $\underline{\hspace{1cm}}$
 $z = 4$
 Point on the plane is $(-5, 0, 4)$ and normal vector of required plane is



$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 3 & -5 & -7 \end{vmatrix} = -7\hat{i} + 7\hat{j} - 8\hat{k}$$

Hence, equation of plane is $-7x + 7y - 8z - 3 = 0$

Perpendicular distance is $\frac{3}{\sqrt{162}} = \frac{1}{3\sqrt{2}}$

4.

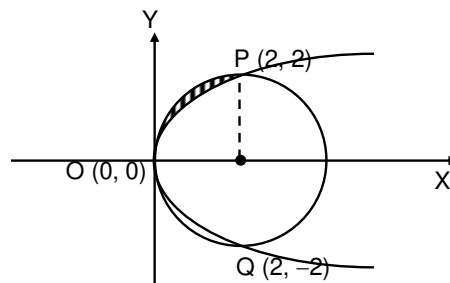
A

The point of intersection of the curve $x^2 + y^2 = 4x$
 $y^2 = 2x$ are $(0, 0)$ and $(2, 2)$ for $x \geq 0$ and $y \geq 0$

So required area = $\frac{1}{4}\pi \times 4 - \int_0^2 \sqrt{2x} dx$

= $\pi - \sqrt{2} \cdot \frac{2}{3} [x^{3/2}]_0^2$

= $\pi - \frac{8}{3}$



5.

C

For non-zero solution $D = 0$

$$\begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 4 & -3 \end{vmatrix} = 0$$

$\Rightarrow 1(-3k + 8) - k(-9 + 4) + 3(12 - 2k) = 0$

$\Rightarrow -3k + 8 + 5k + 36 - 6k = 0$

$\Rightarrow -4k + 44 = 0$

$k = 11$

hence equations are $x + 11y + 3z = 0$

$3x + 11y - 2z = 0$

and $2x + 4y - 3z = 0$

let $z = t$

then we get, $x = \frac{5}{2}t$ and $y = -\frac{t}{2}$

thus, $\frac{xz}{y^2} = \frac{\left(\frac{5}{2}t\right)(t)}{\frac{t^2}{4}} = 10$

6.

B

$\int_{2000}^P dP = \int_0^{25} (100 - 12\sqrt{x}) dx$

$(P - 2000) = 25 \times 100 - \frac{12 \times 2}{3} (25)^{3/2}$

$P = 3500$.

7.

C

Each entry of A is integer, so the cofactor of every entry is an integer and hence each entry in the adjoint of matrix A is integer.

Now $\det A = \pm 1$ and $A^{-1} = \frac{1}{\det(A)} (\text{adj } A)$

$\Rightarrow$ all entries in A^{-1} are integers.

8.

C

Let us assume that 'A' wins after n deuces, $n \in [0, \infty)$

Probability of a deuce = $\frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} = \frac{5}{9}$

(A wins his serve then B wins his serve or A loses his serve then B also loses his serve)

Now probability of 'A' winning the game

$$= \sum_{n=0}^{\infty} (5/9)^n \cdot \left(\frac{2}{3}\right) \frac{1}{3} = \frac{1}{1-(5/9)} \cdot \frac{2}{9} = \frac{1}{2}.$$

9. **A**

$a_1a_1 + b_1b_2 + c_1c_2 = 6 + 2 + 12$ is positive

so that the acute angle bisector is $\frac{2x - y + 2z + 3}{3} = -\frac{3x - 2y + 6z + 8}{7}$

$$\Rightarrow 23x - 13y + 32z + 45 = 0.$$

10. **C**

$$x - 2)^2 + (y - 3)^2 + (z - 4)^2 = r^2$$

Put (0, 1, 0)

$$r^2 = 24$$

11. **C**

$$Y = \cos^{-1}(\cos x) = \begin{cases} x, & 0 \leq x \leq \pi \\ 2\pi - x & \pi < x \leq 2\pi \end{cases}$$

$$\text{The required area} = \int_0^{\pi} x dx + \int_{\pi}^{2\pi} (2\pi - x) dx = \pi^2$$

12. **D**

$$\begin{vmatrix} (a^x + a^{-x})^2 & (a^x - a^{-x})^2 & 1 \\ (b^y + b^{-y})^2 & (b^y - b^{-y})^2 & 1 \\ (c^z + c^{-z})^2 & (c^z - c^{-z})^2 & 1 \end{vmatrix}$$

Operating $C_1 \rightarrow C_1 - C_2$

$$\begin{vmatrix} 4a^x a^{-x} & (a^x - a^{-x})^2 & 1 \\ 4b^y b^{-y} & (b^y - b^{-y})^2 & 1 \\ 4c^z c^{-z} & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & (a^x - a^{-x})^2 & 1 \\ 1 & (b^y - b^{-y})^2 & 1 \\ 1 & (c^z - c^{-z})^2 & 1 \end{vmatrix} = 0.$$

13. **C**

$$\text{The required probability is } {}^nC_1 \frac{1}{2} \left(\frac{1}{2}\right)^{n-1} + {}^nC_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{n-3} + \dots + {}^nC_n \left(\frac{1}{2}\right)^n$$

$$= \left(\frac{1}{2}\right)^n ({}^nC_1 + {}^nC_3 + \dots + {}^nC_n) = \frac{1}{2}.$$

14. **B**

Equation of the required parabola is of the form $y^2 = 4a(x - h)$. Differentiating, we have

$$2y \frac{dy}{dx} = 4a \Rightarrow y \frac{dy}{dx} = 2a$$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 + y \frac{d^2y}{dx^2} = 0.$$

15. **D**

$$A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

A is orthogonal $\Rightarrow AA' = I$

$$\Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Comparing the corresponding elements, we have

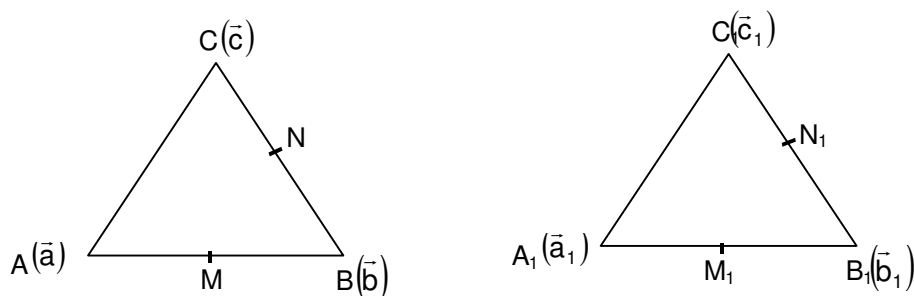
$$4\beta^2 + \gamma^2 = 1$$

$$2\beta^2 - \gamma^2 = 0$$

$$\Rightarrow \beta = \pm \frac{1}{\sqrt{6}}, \gamma = \pm \frac{1}{\sqrt{3}}.$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \Rightarrow \alpha^2 + \frac{1}{6} + \frac{1}{3} = 1 \Rightarrow \alpha = \pm \frac{1}{\sqrt{2}}.$$

16. **D**



$$\overrightarrow{AB} = \overrightarrow{A_1B_1}$$

$$\Rightarrow \vec{b} - \vec{a} = \vec{b}_1 - \vec{a}_1$$

$$\Rightarrow \vec{b} - \vec{b}_1 = \vec{a} - \vec{a}_1 \Rightarrow \overrightarrow{B_1B} = \overrightarrow{A_1A} \Rightarrow \overrightarrow{AA_1} = \overrightarrow{BB_1}$$

$$\Rightarrow \overrightarrow{NN_1} = \frac{\vec{b}_1 + \vec{c}_1 - \vec{b} + \vec{c}}{2} \Rightarrow \overrightarrow{NN_1} = \frac{\vec{b}_1 + \vec{c}_1 - \vec{b} - \vec{c}}{2} \Rightarrow 2\overrightarrow{NN_1} = \overrightarrow{BB_1} + \overrightarrow{CC_1}$$

$$\Rightarrow \overrightarrow{MM_1} = \frac{\vec{b}_1 - \vec{b} + \vec{a}_1 - \vec{a}}{2} \Rightarrow 2\overrightarrow{MM_1} = \overrightarrow{BB_1} + \overrightarrow{AA_1} = 2\overrightarrow{BB_1} = 2\overrightarrow{AA_1}$$

$$\Rightarrow \overrightarrow{MM_1} = \overrightarrow{BB_1} = \overrightarrow{AA_1}.$$

17. **D**

The point $(-1, 2, -2)$ lies on the given line as $\frac{1+1}{2} = \frac{-1-2}{-3} = \frac{2+2}{4}$ is true.

Hence, the length of perpendicular on the given line = 0

18. **A**

$$\begin{aligned} \sum_{r=0}^m \Delta_r &= \begin{vmatrix} \sum_{r=0}^m (2r-1) & \sum_{r=0}^m {}^m C_r & \sum_{r=0}^m 1 \\ m^2 - 1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix} \\ &= \begin{vmatrix} m^2 - 1 & 2^m & m+1 \\ m^2 - 1 & 2^m & m+1 \\ \sin^2(m^2) & \sin^2(m) & \sin(m^2) \end{vmatrix} \\ &= 0 \end{aligned}$$

19. **A**

$$P(C) = 0 \Rightarrow C = \phi \Rightarrow A \cap C = B \cap C = A \cap B \cap C = \phi$$

$$\Rightarrow P(A \cap B \cap C) = 0 = P(A).P(B).P(C) \text{ and } P(A \cap C) = P(B \cap C) = 0 \Rightarrow P(A).(C) = P(B).(C).$$

Hence events are independent.

20.

A

Differential equation can be rewritten as

$$\left(\frac{xdy - ydx}{x^2} \right) \left(\frac{x^2}{y^2} \right) = (x dy + y dx) \sin xy$$

$$\text{or, } \left(d \left(\frac{y}{x} \right) \right) \left(\frac{x^2}{y^2} \right) = (d(xy)) \sin xy$$

Integrating both sides, we get

$$-\frac{1}{y/x} = -\cos(xy) + c \Rightarrow \frac{x}{y} = \cos(xy) - c$$

Put $x = 0, y = 1$, we get $c = 1$.

Part – B

1.

4

$$\sqrt{a^2 - 4} \tan A + a \tan B + \sqrt{a^2 + 4} \tan C = 6a$$

$$\Rightarrow (\sqrt{a^2 - 4}\hat{i} + a\hat{j} + \sqrt{a^2 + 4}\hat{k}) \cdot (\tan A\hat{i} + \tan B\hat{j} + \tan C\hat{k}) = 6a$$

$$\Rightarrow \sqrt{(a^2 - 4) + a^2 + (a^2 + 4)} \sqrt{\tan^2 A + \tan^2 B + \tan^2 C} \cos \theta = 6a$$

where θ is the angle between two vectors. (Since $\vec{p} \cdot \vec{q} = |\vec{p}||\vec{q}| \cos \theta$)

$$\Rightarrow (\sqrt{3}a) \sqrt{\tan^2 A + \tan^2 B + \tan^2 C} = 6a \sec \theta$$

$$= (3a^2) (\tan^2 A + \tan^2 B + \tan^2 C) = 36a^2 \sec^2 \theta$$

$$= \tan^2 A + \tan^2 B + \tan^2 C = 12 \sec^2 \theta \geq 12 \quad (\text{Since } \sec^2 \theta \geq 1)$$

2.

2

The parametric form of the given equation is $x = t, y = t^2$. The equation of any tangent at t is $2xt = y + t^2$. Differentiating we get $2t = y_1 \left(= \frac{dy}{dx} \right)$. Putting this value in the above equation, we have $2x \frac{y_1}{2} = y +$

$$\left(\frac{y_1}{2} \right)^2 \Rightarrow 4xy_1 = 4y + y_1^2.$$

The degree of this equation is 2.

3.

9

The area bounded by the lines

$y = 3 - |x|, -3 \leq x \leq 3$ is shown in the fig.

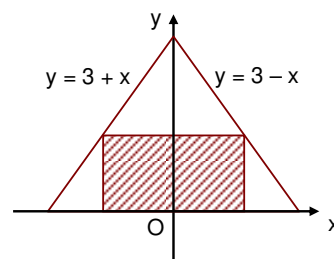
$$\text{Area } A(x) = 2x \cdot (3 - x)$$

$$\Rightarrow A'(x) = 2(3 - x) - 2x$$

$$= 6 - 4x = 0 \Rightarrow x = 3/2$$

$\Rightarrow$ Maximum area of the rectangle occurs when $x = 3/2$.

$$\text{Maximum area} = 2 \cdot \frac{3}{2} \left(3 - \frac{3}{2} \right) = \frac{9}{2} \text{ sq. units.}$$



4.

4

Required area

$$= \int_{\pi/4}^{3\pi/4} (\sin^2 x - \cos^2 x) dx = 1$$

5.

3

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc = 0 \Rightarrow \text{whether } ad = 1, bc = 1 \text{ or } ad = -1, bc = -1$$

which occur in eight ways. Total number of 2×2 determinants from $\{-1, 1\}$ is 16. Thus required probability is $\frac{8}{16} = \frac{1}{2}$.